STAT405 Bayesian Project Proposal

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Project Overview

Our team believes that housing prices are an important factor to consider, especially when it comes to pricing. Moreover, different characteristics of housing size, number of bedrooms, and parking tend to influence the price outcomes. We are interested in how much these variables influence the housing price.

Although most research has already shown that these variables have a major correlation with the housing price. Our team is most interested in measuring the posterior uncertainty of these effects. We want to quantify these types of uncertainty to determine which house features drive the housing prices.

By using publicly available datasets, we will build Bayesian inference models to study how these housing characteristics affect the housing price and estimate the uncertainty of the effect.

Research Question

How do house characteristics such as size, number of bedrooms, and parking affect housing prices, and what is the posterior uncertainty of these effects?

Main dataset

Dataset source:
https://www.kaggle.com/datasets/saurabhbadole/housing-price-data

Description:
The following dataset contains 545 observations and 13 variables and describes various features of each house. The dataset is collected from real estate databases, property listings and other various records.

price	area	bedrooms	bathrooms	stories	mainroad	guestroom	basement	hotwaterheating	airconditioning	parking	prefarea	furnishingstatus
13300000	7420	4	2	3	yes	no	no	no	yes	2	yes	furnished
12250000	8960	4	4	4	yes	no	no	no	yes	3	no	furnished
12250000	9960	3	2	2	yes	no	yes	no	no	2	yes	semi-furnished

12215000	7500	4	2	2	yes	no	yes	no	yes	3	yes	furnished
11410000	7420	4	1	2	yes	yes	yes	no	yes	2	no	furnished
10850000	7500	3	3	1	yes	no	yes	no	yes	2	yes	semi-furnished

Variables included:

- Price - Sales price of the house
- Area - size of the living area in sqft
- Bedrooms - number of bedrooms
- Bathrooms - number of bathrooms
- Stories - number of floors
- Main road - Whether the house is located on the main road (binary - yes/no)
- Guestroom - Is there a guestroom (binary - yes/no)
- Basement - Is there a basement (binary - yes/no)
- Hot water heating - Does the house have hot water heating (binary - yes/no)
- Air conditioning - Is there air conditioning (binary - yes/no)
- Parking - number of parking spaces
- Pref area - is the house located in the preferred residential area (binary - yes/no)
- Furnishing status - Furnishing condition of the house (furnished, semi-furnished, or unfurnished)

Backup dataset

Dataset source:
https://www.kaggle.com/datasets/sukhmandeepsinghbrar/housing-price-dataset/data

Description:

For the backup dataset, we will use another publicly available housing dataset that also contains house prices with some other housing characteristics, such as price, bedrooms and waterfront. The dataset contains around 21 variables and 21614 observations. According to the author, this data was gathered from multiple random real estate websites. The data set has gathered data ranging from 2014 May to 2015 December.

id	date	price	bedrooms	bathrooms	sqft_living	sqft_lot	floors	waterfront	view	condition	grade	sqft_above	sqft_basement	yr_built	yr_renovated	zipcode	lat	long	sqft_living15	sqft_lot15
7.23E+09	20141013T000000	231300	2	1	1180	5650	1	0	0	3	7	1180	0	1955	0	98178	47.5112	-122.257	1340	5650
6.41E+09	20141209T000000	538000	3	2.25	2570	7242	2	0	0	3	7	2170	400	1951	1991	98125	47.721	-122.319	1690	7639
5.63E+09	20150225T000000	180000	2	1	770	10000	1	0	0	3	6	770	0	1933	0	98028	47.7379	-122.233	2720	8062
2.49E+09	20141209T000000	604000	4	3	1960	5000	1	0	0	5	7	1050	910	1965	0	98136	47.5208	-122.393	1360	5000
1.95E+09	20150218T000000	510000	3	2	1680	8080	1	0	0	3	8	1680	0	1987	0	98074	47.6168	-122.045	1800	7503
7.24E+09	20140512T000000	1225000	4	4.5	5420	101930	1	0	0	3	11	3890	1530	2001	0	98053	47.6561	-122.005	4760	101930

Variables included:

- Id – Unique identifier for each house
- Date – Date when the house was sold
- Price – Sale price of the house (response variable)
- Bedrooms – Number of bedrooms in the house
- Bathrooms – Number of bathrooms in the house
- Sqft_living – Total interior living area of the house in square feet
- Sqft_lot – Total land area of the property in square feet
- Floors – Number of floors in the house
- Waterfront – Whether the house has a waterfront view (binary - 0 = no, 1 = yes)
- View – Quality of the view from the house (rating scale)
- Condition – condition of the house (rating scale)
- Grade – construction and design quality of the house
- Sqft_above – Square footage of the house above ground level
- Sqft_basement – Square footage of the basement area
- Yr_built – Year the house was originally built
- Yr_renovated – Year the house was renovated (0 = never renovated)
- Zipcode – Zip code of the property location
- Lat – Latitude coordinate of the house location
- Long – Longitude coordinate of the house location

- Sqft_living15 – Average living area of nearby houses
- Sqft_lot15 – Average lot size of nearby houses

Methodology

Model 1: Baseline Bayesian Normal Regression

With housing price being a continuous variable, we will use Bayesian normal regression:

$$price_i \sim N(\mu_i, \sigma)$$

$$\mu_i = \beta_0 + \beta_1 size_i + \beta_2 bedrooms_i + \beta_3 bathrooms \ \beta_4 stories_i + \beta_5 parking_i$$

We will place weakly informative priors on the regression coefficients for now, for example

$$B_j \sim N(0, 10)$$

and a prior on the positive scale parameter σ.

This model assumes that the variance of housing prices is constant across observations, and we will not assume a standard normal distribution since we consider it to be unrealistic. Thus, the standard deviation is assumed to be 10, but this may be adjusted further.

Model 2: Heteroscedastic Bayesian Normal Regression

A natural extension to allow the variance to depend on predictors.

$$price_i \sim N(\mu_i, \sigma_i)$$

$$\mu_i = \beta_0 + \beta_1 size_i + \beta_2 bedrooms_i + \beta_3 bathrooms + \beta_4 stories_i + \beta_5 parking_i$$

$$log(\sigma_i) = a_0 + a_1 size_i$$

This second model allows the spread of housing prices to vary, which may be realistic if larger or more expensive houses show greater variability. The log makes sure that the variance parameter stays positive, which would allow the predictors to affect the scale of variability.

Alternative second model

Hierarchical normal regression would be considered with a more detailed dataset of housing areas, since we don't have neighbourhoods or districts, but we do have zip codes which could make this model possible, but we are still in discussion of how to clean the dataset further to make the zip codes usable.

Methods for Computation

Method 1: Laplace Approximation

We believe laplace approximation would be suitable for this topic. By estimating maximum a posteriori, we can easily find the Gaussian approximation of the posterior. Given the model we have provided from the normal regression and the heteroscedastic normal regression, we can find the approximation of the posterior around the mode and the curvature of the log posterior.

Method 2: Metropolis–Hastings

We will use MH to check if the Laplace approximation is valid, as we can't automatically assume asymptotic normality. Thus, by using MH and comparing the results, we can assess whether this approximation adequately captures posterior uncertainty or not. Moreover, MH allows us to generate samples from the posterior distribution over many iterations, which then can be used to estimate posterior summaries.

Goal of the model/posterior methods

For the two models, we will compare:

- Posterior estimates of regression coefficients
- Posterior uncertainty for the effects of living area size, number of bedrooms, and number of bathrooms, etc
- Predictive performance
- Goodness of fit

For the two posterior methods, we will compare:

- Posterior means
- Posterior credible intervals
- Whether Laplace and MH give similar conclusions about the effects of the predictors

MCMC diagnostics

For Metropolis–Hastings, we will examine:

- trace plots
- acceptance behavior
- posterior histograms
- effective sample size if available
- whether multiple chains mix similarly

Work distribution

One of us will work on the literature review while the other will focus on the setups for the data exploration and model design. One member will focus primarily on implementing the Laplace approximation, while the other will implement the Metropolis–Hastings algorithm. After Quiz 2, we will expand the methods as we will have more knowledge on MCMC. For now, we will implement the 2 methods mentioned since it's the best available option.

All project materials will be stored in a shared GitHub repository, allowing both members to contribute code and keep track of each other's code.

Repository

https://github.com/EdmundChuu/STAT-405-Bayesian-Statistics-Project

Overall structure on GitHub:

- proposal/ – project proposal
- data/ – datasets used in the analysis
- code/ – scripts implementing Bayesian models and inference
- report/ – final report and figures